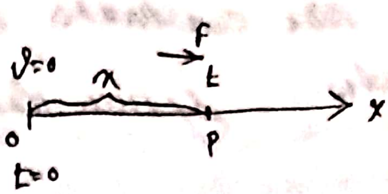
§ Dynamics of a particle §

Motion in a st. line

§ Equation of motion :-

Let O be the origin and a particle start from O and move along OX. Let P be the position of the particle at any time t, such that OP = x. Let v be the velocity of the particle at P. Then $v = \frac{dx}{dt}$ = Velocity of the particle at P [in the direction of x-axis]



Now $f = \frac{d^2x}{dt^2}$ = Acceleration of the particle at P

$= \frac{dv}{dt}$ (in the direction of x increasing)

$= v \cdot \frac{dv}{dx}$

Then equation of motion of the particle of mass m is

$m\ddot{x} = F$ [F being the external force acting on the particle in the positive direction of x-axis]

or

$m\ddot{x} = -F$ [F being the external force acting on the particle in negative direction of x axis]

§ Important Relations :-

1. $v = u + ft$, u being the initial velocity.
2. $x = ut + \frac{1}{2}ft^2$
3. $v^2 = u^2 + 2fx$

§ Prob. 1:- If the velocity v of a particle moving in a st. line is given by $v^2 = ax^2 + b$, where x is the distance travelled from a fixed point on the line and a, b are constants, the acceleration varies as ——

Ans. given that $v^2 = ax^2 + b$

D.w.r. to x

$$2v\frac{dv}{dx} = 2ax$$

$$\text{or } f = ax \quad \left[\text{as } v\frac{dv}{dx} = f = \text{acceleration}\right]$$

$$\text{or } f \propto x \text{, } a \text{ being the constant.}$$

§ Prob. 2:- A particle moves along the st. line according to the law $s^2 = 6t^2 + 4t + 3$, where s is the distance of the particle at time t. Then the velocity of the particle at time $t = 1$ is equal to ——

Ans. given that $s^2 = 6t^2 + 4t + 3$

D.w.r. to t

$$2s.\frac{ds}{dt} = 6.2.t + 4$$

$$\text{or } \frac{ds}{dt} = \frac{6t + 2}{\sqrt{6t^2 + 4t + 3}}$$

$$\text{or } \left.\frac{ds}{dt}\right|_{t=1} = \frac{6.1 + 2}{\sqrt{6.1^2 + 4.1 + 3}} = \frac{8}{\sqrt{13}}$$

$$\text{or } \left.\frac{ds}{dt}\right|_{t=1} = \frac{8}{\sqrt{13}}$$

§ Prob. 3:- The velocity of a particle moving in a st. line given by the relation $v^2 = ax^2 + bx + c$, where v is the velocity of the particle and x is the distance of the particle from fixed point at time t. The acceleration of the particle is ——

Ans. given that $v^2 = ax^2 + bx + c$

D.w.r. to x

$$2v\frac{dv}{dx} = 2ax + b$$

$$\text{or, } \boxed{v\frac{dv}{dx} = ax + b/2}$$

§ Prob. 4:- Which of the following is scalar?

i) displacement ii) Speed iii) Velocity iv) Acceleration

Ans. Speed is a scalar quantity. It has magnitude but does not have direction.

§ Prob. 5:- A particle moves along a st. line according to the law $s^2 = 6t^2 + 4t + 3$, where s is the displacement and t time, then its acceleration varies as ——

Ans. Given that $s^2 = 6t^2 + 4t + 3$

D. w. r. to t

$$2s\dot{s} = 12t + 4$$

$$\text{or } s\ddot{s} + \dot{s}^2 = 6$$

$$\text{or, } s\ddot{s} + \left(\frac{6t+2}{s}\right)^2 = 6$$

$$\text{or, } \ddot{s} + \frac{4(9t^2 + 6t + 1)}{s^3} = 6/s$$

$$\text{or } \ddot{s} = 6/s - \frac{4(9t^2+6t+1)}{s^3}$$

$$= \frac{6s^2 - 36t^2 - 24t - 4}{s^3}$$

$$= \frac{36t^2 + 24t + 18 - 36t^2 - 24t - 4}{s^3}$$

$$\text{or } \ddot{s} = \frac{14}{s^3}$$

$$\text{or } \ddot{s} \propto \frac{1}{s^3}$$

$$\therefore \boxed{f \propto \frac{1}{s^3}}$$

§ Prob. 5:- The law of motion of a particle moving in a st. line is $s = \frac{1}{2}vt$. Then the acceleration is ——

Ans. given that $s = \frac{1}{2}vt$

D. w. r. to t

$$\dot{s} = \frac{1}{2}\left[t\frac{dv}{dt} + v\right]$$

$$\text{or, } 2v - v = \frac{dv}{dt} \cdot t$$

or $v = f \cdot t$

D.w.r. to t

$$\frac{dv}{dt} = f \cdot 1 + t \cdot \frac{df}{dt}$$

or, $t \cdot \frac{df}{dt} = 0$

$\Rightarrow t = 0$ Which is impossible. / $\frac{df}{dt} = 0$ $\Rightarrow f$ is constant.

§ Prob. 6:- If the time t be regarded as function of velocity v, then, the rate of decrease of acceleration f is ____

Ans. given that $t = g(v)$

Now $\frac{df}{dt} = \dfrac{\frac{d}{dt}\left(\frac{dv}{dt}\right)}{}$ [By chain Rule]

$$= \frac{d}{dv}\left(\frac{dv}{dt}\right) \cdot \frac{dv}{dt}$$

$$= \frac{d}{dv}\left(\frac{\frac{dv}{dv}}{\frac{dt}{dv}}\right) \cdot f$$

$$= \frac{d}{dv}\left(\frac{1}{\frac{dt}{dv}}\right) \cdot f$$

$$= f \cdot -\frac{1}{\left(\frac{dt}{dv}\right)^2} \cdot \frac{d^2t}{dv^2}$$

$$\boxed{\frac{df}{dt} = -f^3 \cdot \frac{d^2t}{dv^2}}$$

§ Prob. 7:- A point moves in a st. line so that its distance s from a fixed point at time t is proportional to t^n. If v be the velocity & f the acceleration at any time t, show that $\boxed{(n-1)v^2 = nfs}$

Ans. given that $s \propto t^n$

or, $s = kt^n$, where k is the constant of proportionality.

Then $\frac{ds}{dt} = k n \cdot t^{n-1}$ —— ①

And $\frac{d^2s}{dt^2} = k n(n-1) \cdot t^{n-2}$

Now, $L.H.S = nfs = n \cdot \frac{d^2s}{dt^2} \cdot s$

$= n \cdot k \cdot n(n-1) \cdot t^{n-2} \cdot k \cdot t^n$

$= (n-1) k^2 \cdot n^2 \cdot (t^{n-1})^2$

$= (n-1) \cdot v^2$

$= R.H.S$

$\therefore \boxed{L.H.S = R.H.S}$

§ Prob. 8:- A particle is moving in a st. line, the distance x from a fixed point on its path at time t is given by $t = ax^2 + bx + c$, where a, b, c are constants. Find the velocity v in terms of x and show that the retardation of the particle is $2av^3$.

Ans given that $t = ax^2 + bx + c$

D. w. r. to x

$\frac{dt}{dx} = 2ax + b$

or $\frac{1}{\frac{dx}{dt}} = 2ax + b$

or $\frac{1}{v} = 2ax + b$

or, $v = \frac{1}{2ax+b}$

Now acceleration $= v \frac{dv}{dx}$

$= \frac{1}{(2ax+b)} \cdot \frac{d}{dx}\left(\frac{1}{2ax+b}\right)$

$= \frac{1}{(2ax+b)} \cdot -\frac{1}{(2ax+b)^2} \cdot 2a$

$\boxed{f = -2av^3}$ $\quad v = \frac{1}{2ax+b}$

§ Prob. 7:- A particle moves towards a centre of attraction starting from rest at a distance a from the centre; if its velocity when at any distance x from the centre vary as $\sqrt{\frac{a^2-x^2}{x^2}}$, find the law force.

Ans. given that $\frac{dx}{dt} = -\sqrt{\frac{a^2-x^2}{x^2}}$ [Negative sign is taken as t increases x decreases]

or $\left(\frac{dx}{dt}\right)^2 = \left(-\sqrt{\frac{a^2-x^2}{x^2}}\right)^2$

or $\left(\frac{dx}{dt}\right)^2 = \frac{a^2-x^2}{x^2} = \frac{a^2}{x^2} - 1$

~~d~~ D. w. r. to ~~x~~ t

$2 \cdot \frac{dx}{dt} \cdot \frac{d^2x}{dt^2} = -2a^2 \cdot \frac{1}{x^3} \cdot \frac{dx}{dt}$

or $\boxed{\frac{d^2x}{dt^2} = -\frac{a^2}{x^3}}$

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